

Miracle Advance Technologies

DIGITAL EXPERIENCE HEALTH CHECK

Application User Manual

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1.1. Organisation of the Manual

The manual is designed to initially familiarize you with the application's most common features, such as dashboards and the search feature. While doing so, the most frequent elements are explained so that you can later relate to the terms as you explore the application's features and functionalities in greater depth. Items in the Main Menu will vary depending on user responsibilities, but the complete menu is studied as the manual progress, and user roles are explained ahead of time, so the menu options that each user position sees may be understood. The Study Worklist includes the most fundamental and necessary patient details that the doctors perform. Best efforts were made to write the features in a way that follows the chronology used by the majority of user roles in the system. This section delves deeply into the hospital's multiple operations and management of inpatients. The documentation advances in this part based on the submenu within the Digital Experience Health Check application. Following that, the sections that follow will

highlight the various tools and capabilities within the application that you may find more useful once you've gotten to know your way around Digital Experience Health Check.

1.2. Additional Help

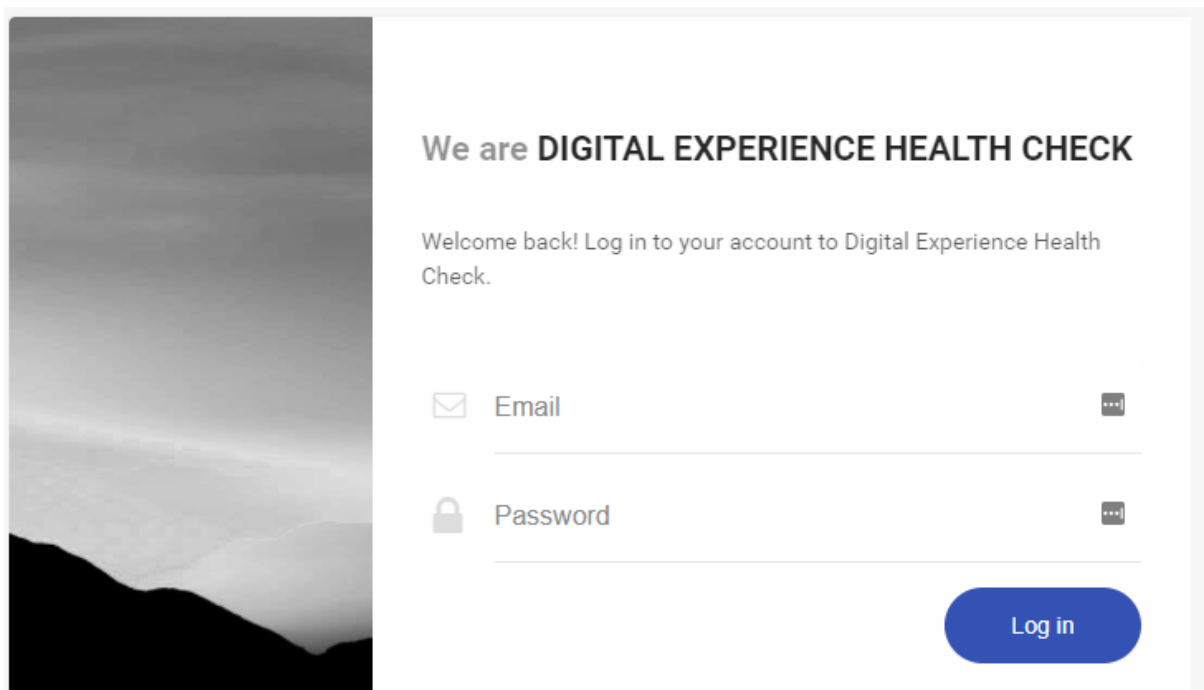
Throughout the user manual, common terms are bolded to familiarize the reader with the terms used in the application. Section numbers are indexed (numbered) and/or linked next to popular keywords and task phrases, allowing the reader to go directly to the section for more information as needed. Keywords in blue are linked for easier navigation, so you may return to the part where you can get specifics about keywords/phrases.

1.3. Browser Requirements

- Latest Chrome browser
- Edge
- Latest Mozilla Firefox
- Internet Explorer
- Safari

1.4. Getting Started

Before you can log in to Digital Experience Health Check, you will need to have your email registered by the System Administrator to be a user of the system. Go to <https://doc.dinsight.health/> and log in with your credentials. The login page should look like this



We are DIGITAL EXPERIENCE HEALTH CHECK

Welcome back! Log in to your account to Digital Experience Health Check.

Email

Password

Log in

Once logged in, you should be viewing the **Study Worklist**. From here you will find the data of this patient.

1.5. Study Worklist

From this Study Worklist, the user will find different options to operate the data.

Study Worklist : Data Center

From Date : 03-12-2021 To Date : 03-12-2021 Hospital : 92 items selected Group : 137 items selected Search Today Yesterday Last 3 Day Last 7 Day

Import Studies Add to PACS Queue Appointment List Appointment Card Finalized Report Scan Accession#

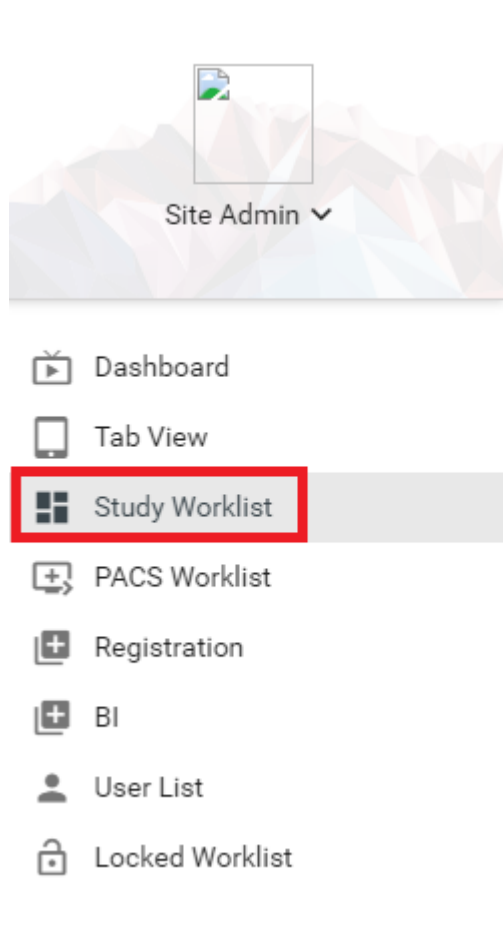
☐ Completed ☐ Finalized ☒ Completed & Finalized ☐ Scheduled/In-progress

Show All Auto Refresh Selected Study(s) : 0 / 161

14 columns selected CSV Export : All CSV Export : Selection Only

☐	Queue#	Access	HN	Patient Name	Date of Birth	Gender	Exam Name	Image	Status	Finalized Category	Compared	Created On	Abnormality	Lung Opacity	Action
☐	21120	ISO5	ISO5	รศ.ดร.วิภาดา	08-11-2021	Female	CHEST PA UPRIGHT	1	Finalized	Category 1	Stable	03-12-2021 08:59 AM	0.1 %	0.1 %	     

In the left corner, the user will find the **Study Worklist**.



1.6. Understanding the Features

Before you can start to use Digital Experience Health Check, it is imperative that you learn about the small features that are incorporated throughout the application for ease of use, convenience, and better user experience.

The most default feature you will find in almost any tabulated information list is the Column Settings option. Here's how you see it:

From Date : 03-12-2021 To Date : 03-12-2021 Hospital : 92 items selected Group : 135 items selected
 Search Today Yesterday Last 3 Day Last 7 Day
Show All Auto Refresh Selected Study(s) : 1 / 129

Import Studies Add to PACS Queue Appointment List Appointment Card Finalized Report Scan Accession#

☐ Completed ☒ Finalized ☐ Completed & Finalized ☐ Scheduled/In-progress

14 columns selected CSV Export : All CSV Export : Selection Only

Queue#	Technical Quality	Accession#	HN	SSN	HN	Patient Name	Gender	Exam Name	Image	Status	Finalized Category	Compared	Created On	Abnormality	Lung Opacity	Technical Quality	Action
05	IS0523571	วรัทยา ะ ทะโย	Female	CHEST PA UPRIGHT	1	Finalized	Category 1	Stable	03-12- 2021 08:59 AM	0.1 %	0.1 %	A					

Hospital: Here the user can select items from the dropdown.

From Date : 03-12-2021 To Date : 03-12-2021 Hospital : 92 items selected Group : 135 items selected
 Search

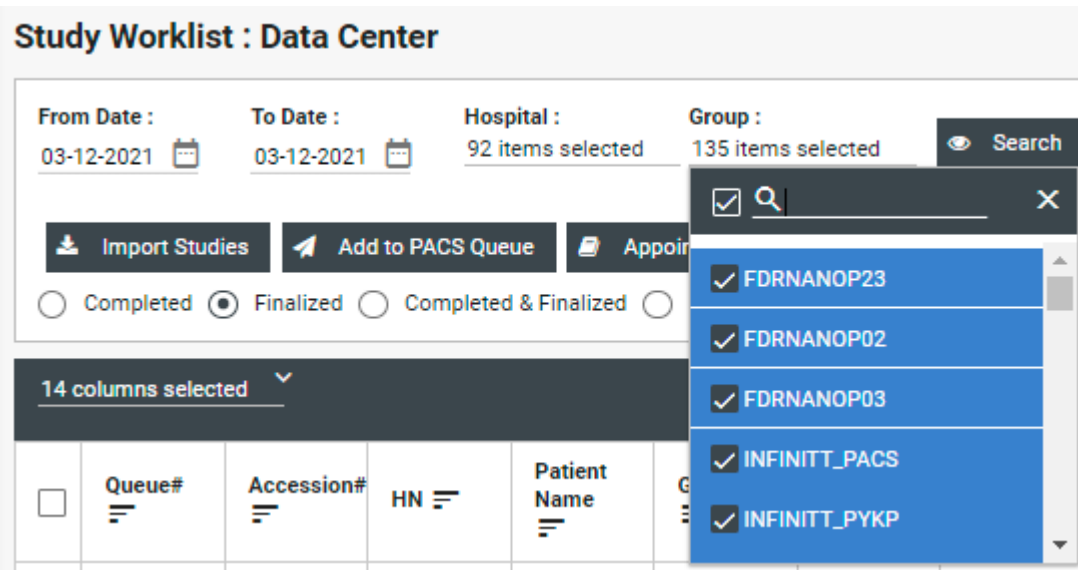
Import Studies Add to PACS Queue Appointment List Appointment Card Finalized Report Scan Accession#

☐ Completed ☒ Finalized ☐ Completed & Finalized ☐ Scheduled/In-progress

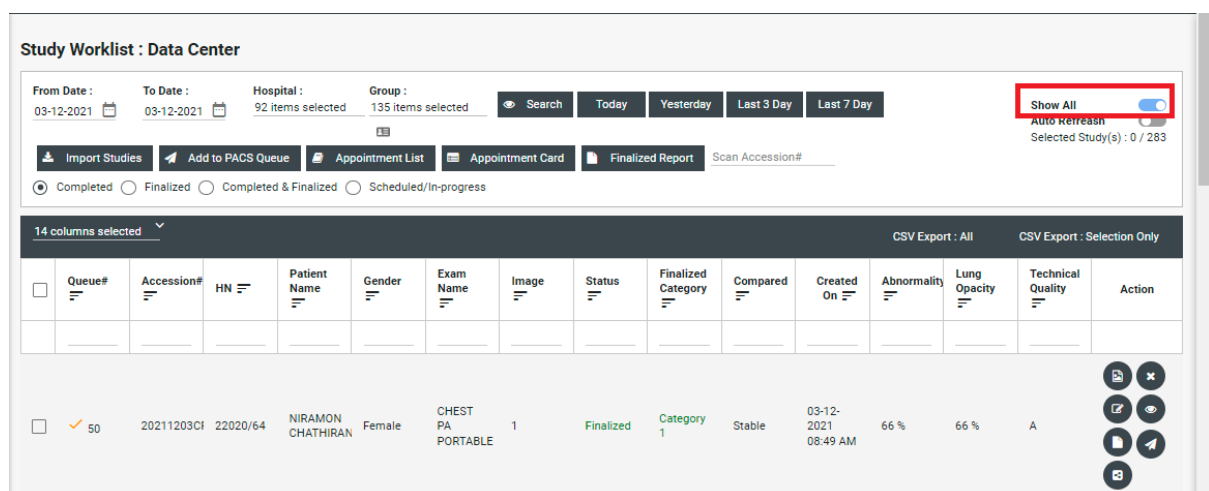
14 columns selected

☒ 1.เรือนจำอำเภอธัญบุรี
☒ 2.เรือนจำพิเศษกรุงเทพมหานคร
☒ 3.เรือนจำกลางคลองเปรม
☒ 8.โรงพยาบาลสนาม มหาสารคาม
☒ 9.ทัณฑสถานบำบัดพิเศษกลาง กรุงเทพมหานคร

Group: Here the user can select group items from the dropdown.

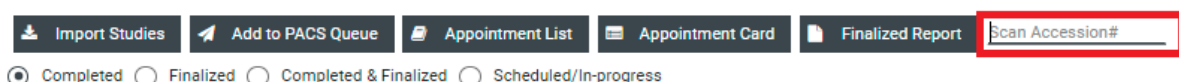


Show All: The user can find all types of status by simply clicking to Show all toggle button in the right corner of the application.



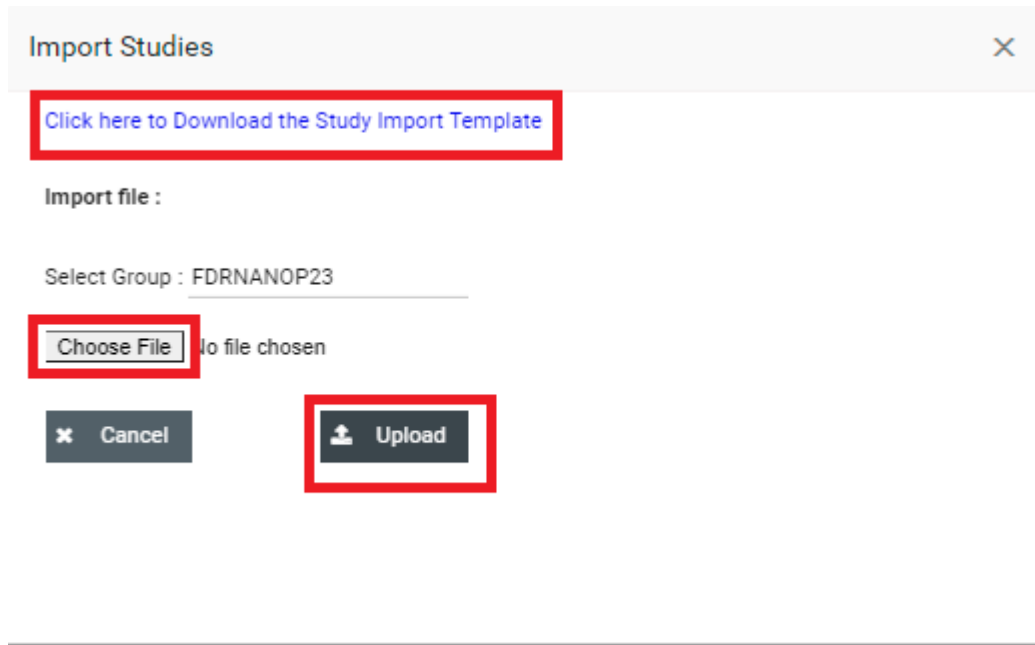
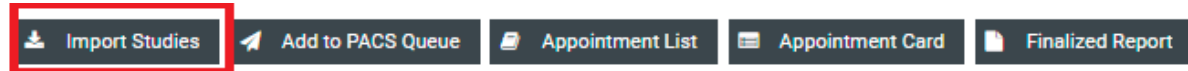
Scan Accession:

The user can scan the accession right after the registration for showing the arrival schedule of the patient.



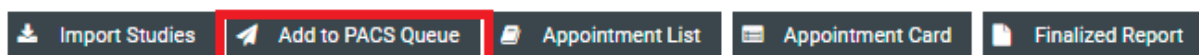
Import Studies

The user can import studies and also download the study Import templates.

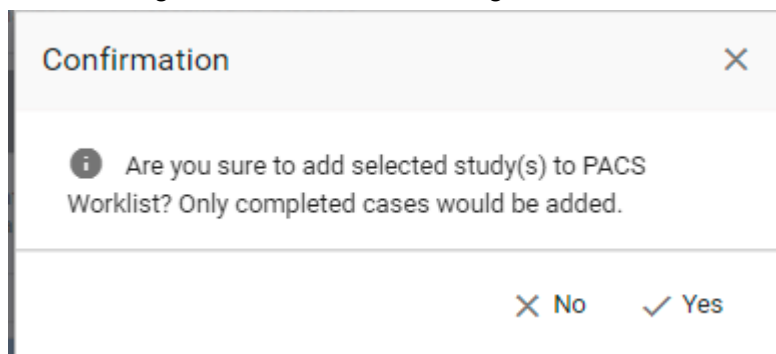
A dialog box titled 'Import Studies' with a close button (X) in the top right corner. Inside the dialog, there is a blue hyperlink 'Click here to Download the Study Import Template' which is highlighted with a red box. Below the link, the text 'Import file :' is followed by a text input field containing 'Select Group : FDRNANOP23'. Below the input field is a 'Choose File' button, also highlighted with a red box, followed by the text 'No file chosen'. At the bottom of the dialog, there are two buttons: 'Cancel' and 'Upload', both highlighted with red boxes.

Add to PACS Queue

The user can add a selected study to **PACS Worklist**.



After clicking the button, the user will get a confirmation notification.

A confirmation dialog box titled 'Confirmation' with a close button (X) in the top right corner. The dialog contains an information icon (i) followed by the text: 'Are you sure to add selected study(s) to PACS Worklist? Only completed cases would be added.' At the bottom right of the dialog, there are two buttons: 'No' and 'Yes', both preceded by a checkmark icon.

Finalized Report

The user can see the finalized report by selecting any finalized status.

Import Studies

Add to PACS Queue

Appointment List

Appointment Card

Finalized Report

Scan Accession#

Completed

Finalized

Completed & Finalized

Scheduled/In-progress

14 columns selected

CSV Export : All

CSV Export : Selection Only

☐	Queue#	Accession#	HN	Patient Name	Date of Birth	Gender	Exam Name	Image	Status	Finalized Category	Compared	Created On	Abnormality	Lung Opacity	Action
☒	✓ 211206IS05	ISO523599		ชุตินันท์ วิเศษธรรม สาธิต	15-01-1977	Male	CHEST PA UPRIGHT	1	Finalized	Category 1		06-12-2021 07:56 AM	56 %	0.1 %	

FINALIZED REPORT

First Name : ชุตินันท์	Study : CHEST PA UPRIGHT
Last Name : วิเศษธรรมสาธิต	SSN : 3909800050678
Age : 44.00	HN : IS0523599
Gender : Male	Finalized By : RCRT Team
	Created On : 06.12.2021

Findings :

Lungs : Normal

Pleura : Normal

Heart and Mediastinum: Normal

Others : Normal

Impression :

Category: Category 1 : Normal chest X-ray or no abnormality detected.

DISCLAIMER : Abnormality Category adapted from Rama Co-RADS version 1

AI Findings :

Abnormality Score : 56 %

AI Finding Report :

Tuberculosis : 56%

Atelectasis : 0.1%

Cardiomegaly : 0.1%

Pulmonary Edema : 0.1%

Lung Opacity : 0.1%

Nodule : 0.1%

Mass : 0.1%

Pleural Effusion : 0.1%

Status

The user can select status individually to see the specific column.

Completed

Finalized

Completed & Finalized

Scheduled/In-progress

14 columns selected

CSV Export : All

CSV Export : Selection Only

☐	Queue#	Accession#	HN	Patient Name	Date of Birth	Gender	Exam Name	Image	Status	Finalized Category	Compared	Created On	Abnormality	Lung Opacity	Action
☐	✓	211206IS05	ISO523599	ชุตินันท์ วิเศษธรรม สาธิต	15-01-1977	Male	CHEST PA UPRIGHT	1	Finalized	Category 1		06-12-2021 07:56 AM	56 %	0.1 %	

On the right side of the column, the user will find the Action column.

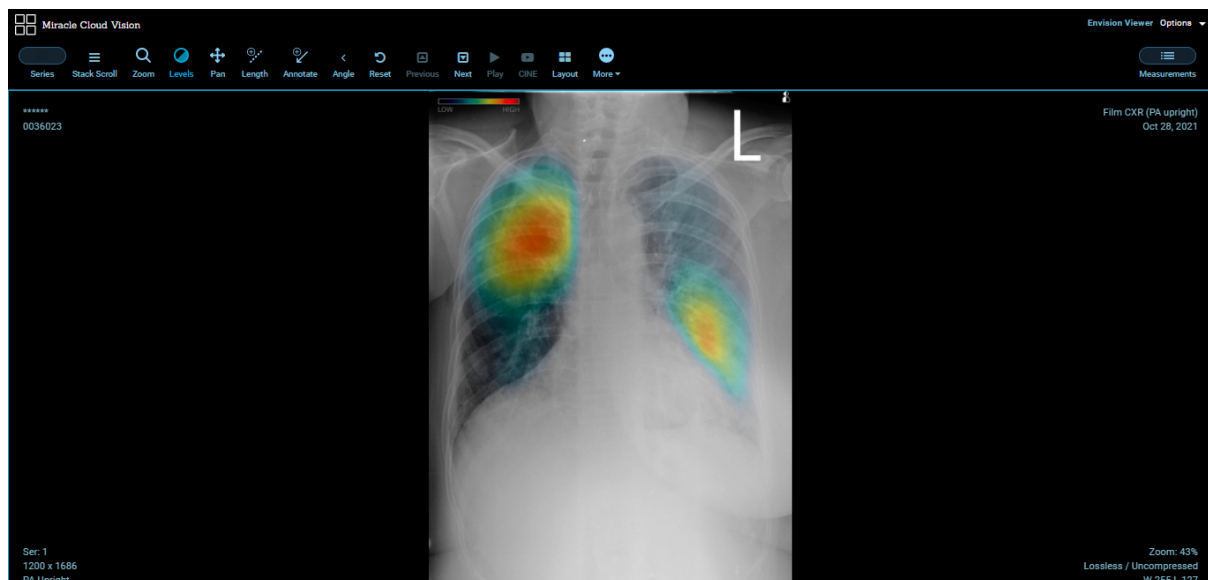
14 columns selected														CSV Export : All		CSV Export : Selection Only	
☐	Queue#	Accession#	HN	Patient Name	Date of Birth	Gender	Exam Name	Image	Status	Finalized Category	Compared	Created On	Abnormality	Lung Opacity	Action		
☐																	
☐	☒	211206IS05	ISO523599	สุวิทย์ รังษานุรักษ์ สกล	15-01-1977	Male	CHEST PA UPRIGHT	1	Finalized	Category 1		06-12-2021 07:56 AM	56 %	0.1 %			

The user will find some buttons in the Action column.

- **View AI Image**
- **Cancel**
- **Edit**
- **View Result**
- **View Report**
- **Resend to RIS**
- **Copy Shareable Link**

View AI Image

The user can see the AI image by clicking the **View AI Image**.



This is a DICOM image, where the AI marks the abnormal area and gives the output.

The user will find the AI findings by clicking the **View Result**.

AI Finding

Abnormality Score : 56 %

AI Finding Report :

Tuberculosis : 56%

Atelectasis : 0.1%

Cardiomegaly : 0.1%

Pulmonary Edema : 0.1%

Lung Opacity : 0.1%

Nodule : 0.1%

Mass : 0.1%

Pleural Effusion : 0.1%

AI DISCLAIMER: This is an AI generated Finding.

DISCLAIMER:

Edit

The user can update Study Info with the edit button inside the Action column.

Update Study Info

Patient Information

SSN :	3909800050678	HN :	ISO523599
First Name :	ชัชวาลย์	Last Name :	จินตนากรสากล
DOB :	15-01-1977	Gender :	Male
Age	44		

Study Information

Queue Serial :	Queue	Accession No :	211206ISO50001
Study Name :*	CHEST PA UPRIGHT	Study Date :*	06-12-2021

Confirm

Confirm & Generate MWL

View Report

The user will view the finalized report form here.

FINALIZED REPORT			
First Name :	วชิร	Study :	CHEST PA UPRIGHT
Last Name :	วิมลวิมลวิมล	SSN :	3909800050678
Age :	44.00	HN :	ISO523599
Gender :	Male	Finalized By :	RCRT Team
		Created On :	06.12.2021

Findings :

Lungs : Normal

Pleura : Normal

Heart and Mediastinum: Normal

Others : Normal

Impression :

Category: Category 1 : Normal chest X-ray or no abnormality detected,

DISCLAIMER : Abnormality Category adapted from Rama Co-RADS version 1

AI Findings :

Abnormality Score : 56 %

AI Finding Report :

Tuberculosis : 56%

Atelectasis : 0.1%

Cardiomegaly : 0.1%

Pulmonary Edema : 0.1%

Lung Opacity : 0.1%

Nodule : 0.1%

Mass : 0.1%

Pleural Effusion : 0.1%

Signature Printed Date: 06.12.2021 Page 1 of 1

When the user doubles click the column, they will find **Reporting** Page and **Envision DICOM Viewer**.

They will first see the patient details and can select the Technical Quality based on AI Findings. Moreover, they can select Template.

Reporting

First Name :	นาง ดอญะณี แก้วสิงห์	Last Name :		Age :	55	Gender :	Female
Accession No :	2470419933	Study Name :	Film CXR (PA upright)	HN :	0036023	SSN :	
Queue Serial :	2	Study Date :	Dec 7, 2021	Image :	1	RT-PCR positive Date :	
Comparison :	☒ No	DICOM Viewer :	Open	Counter Status :	0 : 78 : 130	Status :	Completed

Technical Quality ☐ Good ☒ Acceptable with some technical defect, but still adequate for interpretation. ☐ Unacceptable for interpretation. Repeat chest X-ray is required.	Template: Select Template 📄
---	---

In below, the user will find some options.

- AI/History
- Finding
- Impression

AI/History

Here, the user will find the AI Findings of the patient which is detected by the AI.

The screenshot shows a web interface with three tabs: 'AI/History', 'Finding', and 'Impression'. The 'Finding' tab is active. On the left, a red box highlights the 'Abnormality Score : 95 %' and the 'AI Finding Report :'. The report lists: Tuberculosis : 95%, Atelectasis : 40%, Cardiomegaly : 56%, Pulmonary Edema : 0.1%, Lung Opacity : 92%, Nodule : 62%, Mass : 0.1%, and Pleural Effusion : 0.1%. Below this, another red box highlights the 'AI DISCLAIMER: This is an AI generated Finding.' and a 'DISCLAIMER:' label. On the right, the 'Report :' section shows 'Findings : Lungs : Normal, Pleura : Normal, Heart and Mediastinum: Normal, Others : Normal'. Below this is a 'NORMAL' button. Further down are three buttons: 'Mark as Normal' (green), 'Save as Final' (grey), and 'Skip' (dark grey). At the bottom right is a checkbox for 'Load Next Patient'.

The user can select findings by own if needed. However, AI has already provided the findings.

The screenshot shows the 'Finding' tab. A red box highlights a section with a checked checkbox 'Any significant abnormality?' and three options: 'No', 'Minor abnormalities such as granuloma, mild cardiomegaly, atherosclerotic aorta, minimal apical pleural thickening, plate atelectasis, breast implants, scoliosis, spondylosis and old fracture', and 'Significant abnormalities'. Below this, another red box highlights the 'AI DISCLAIMER: This is an AI generated Finding.' and a 'DISCLAIMER:' label. On the right, the 'Report :' section shows 'Findings : Lungs : Normal, Pleura : Normal, Heart and Mediastinum: Normal, Others : Normal'. Below this is a 'NORMAL' button. Further down are three buttons: 'Mark as Normal' (green), 'Save as Final' (grey), and 'Skip' (dark grey).

The user can select Impression by own if needed. However, AI has already provided the impressions. When the user selected any field the button changed from **Normal** to **Abnormal** and change its color.

The screenshot shows a 'Category' section on the left with five options. Option 3, 'Equivocal for COVID-19 pneumonia.', is selected with a checked checkbox. A red box highlights this section. On the right, the 'Findings :' section shows 'Lungs : Normal, Pleura : Normal, Heart and Mediastinum: Normal, Others : Normal'. Below this is an 'Impression :' section with a red box highlighting an 'ABNORMAL' button. Further down are three buttons: 'Mark as Normal' (green), 'Save as Final' (grey), and 'Skip' (dark grey).

AI/History Finding Impression

☒ Category
☐ Any change as compared with prior study?

Category
☐ 1: Normal chest X-ray or no abnormality detected
☐ 2: Minor abnormalities unrelated to COVID-19 pneumonia.
☐ 3: Low probability of COVID-19 pneumonia, but with other clinically significant diseases.
☐ 3: Equivocal for COVID-19 pneumonia.
☒ 4: Suspicious for early/mild COVID-19 pneumonia.
☐ 5: Suggestive of COVID-19 pneumonia; or follow-up known COVID-19 pneumonia.

AI DISCLAIMER: This is an AI generated Finding.

DISCLAIMER:

Report :

Findings :

Lungs : Normal
Pleura : Normal
Heart and Mediastinum: Normal
Others : Normal

Impression :

Category; Category 4: Suspicious for early/mild COVID-19 pneumonia.

ABNORMAL

☒ Mark as Normal

☒ Save as Final

☒ Skip

☒ Load Next Patient

AI/History Finding Impression

☒ Category
☐ Any change as compared with prior study?

Category
☒ 1: Normal chest X-ray or no abnormality detected
☐ 2: Minor abnormalities unrelated to COVID-19 pneumonia.
☐ 3: Low probability of COVID-19 pneumonia, but with other clinically significant diseases.
☐ 3: Equivocal for COVID-19 pneumonia.
☐ 4: Suspicious for early/mild COVID-19 pneumonia.
☐ 5: Suggestive of COVID-19 pneumonia; or follow-up known COVID-19 pneumonia.

AI DISCLAIMER: This is an AI generated Finding.

DISCLAIMER:

Report :

Findings :

Lungs : Normal
Pleura : Normal
Heart and Mediastinum: Normal
Others : Normal

Impression :

Category; Category 1 : Normal chest X-ray or no abnormality detected.

NORMAL

☒ Mark as Normal

☒ Save as Final

☒ Skip

☒ Load Next Patient

Note: The user can check **Load Next Patient** to get the immediate next patient.

2. PACS Worklist

In PACS Worklist, the user will find the information regarding PACS. Moreover, they can select **Bus** from the dropdown.

PACS Worklist

Bus :

MD-3

Add to PACS Queue

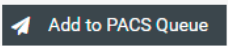
11 columns selected

Global filter

	Queue#	Bus	Accession	HN	Patient Name	Gender	Exam Name	Modality	Image	PACS Status	Created On	Action
☐												

10

The user can also add a selected study to the PACS queue by simply clicking the **Add to**

PACS Queue  button. After clicking the button, the user will get a confirmation pop-up.

Confirmation

Are you sure to add selected study(s) to PACS Worklist? Only completed cases would be added.

No

Yes

3. Registration

In this section, the user can register for the study.

Study Registration

Patient Information

First Name

First Name

Last Name

Last Name

DOB

Gender

Male

Age

Age

HN

HN

☒ Auto Generate

SSN *

SSN

☒ Auto Generate

Study Information

Accession No :

Accession#

☒ Auto Generate

Selected Group *

FDRNANOP91

Queue# *

Queue

Select Study *

CHEST PA UPRIGHT

Save

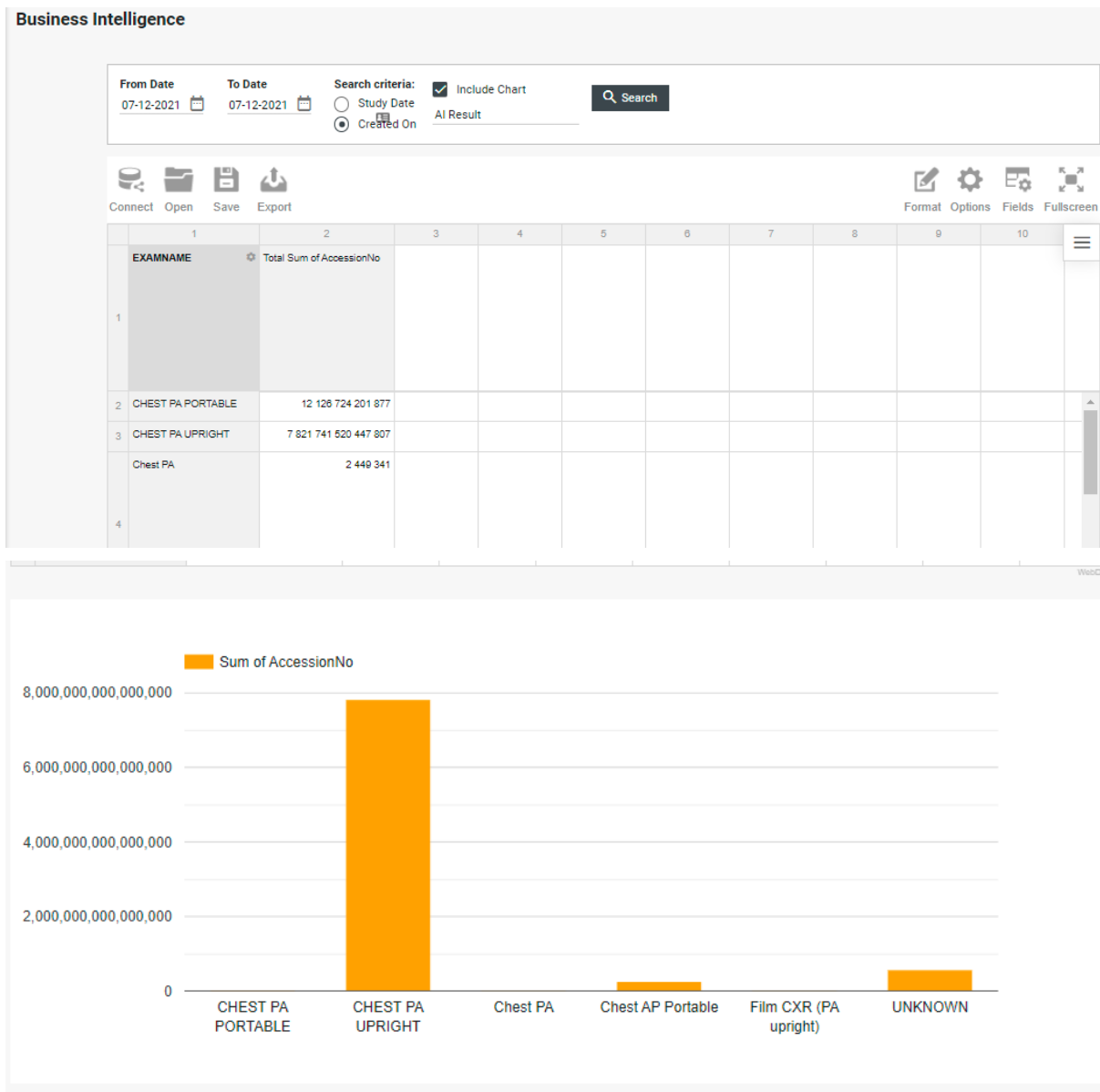
Save & Preview

Note: * Mandatory Field

HN and SSN will be auto-generated when the user clicks the checkbox.

4. BI

In this Business Intelligence section, the user will get raw data and statistical data for better visualization.



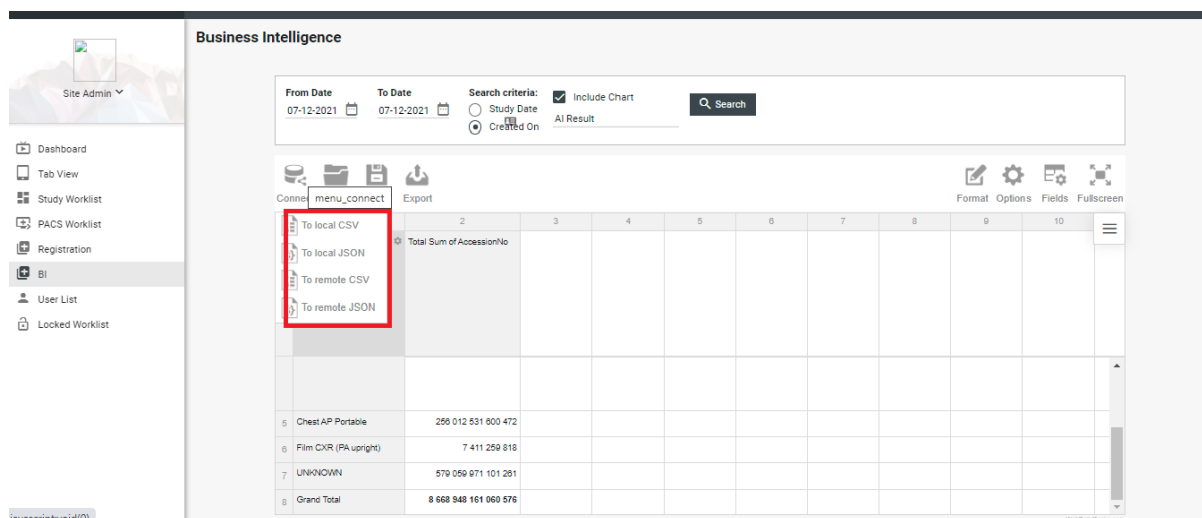
In the first part, the user can search data, From Date -To Date. They can also select whether they want to see **Study Date** and **Created On** from the **Search Criteria**. Also, the user can select the **Include Chart** option to view the visualization of the data.

In the left corner users will find some functions like- Connect, Open, Save, and Export.

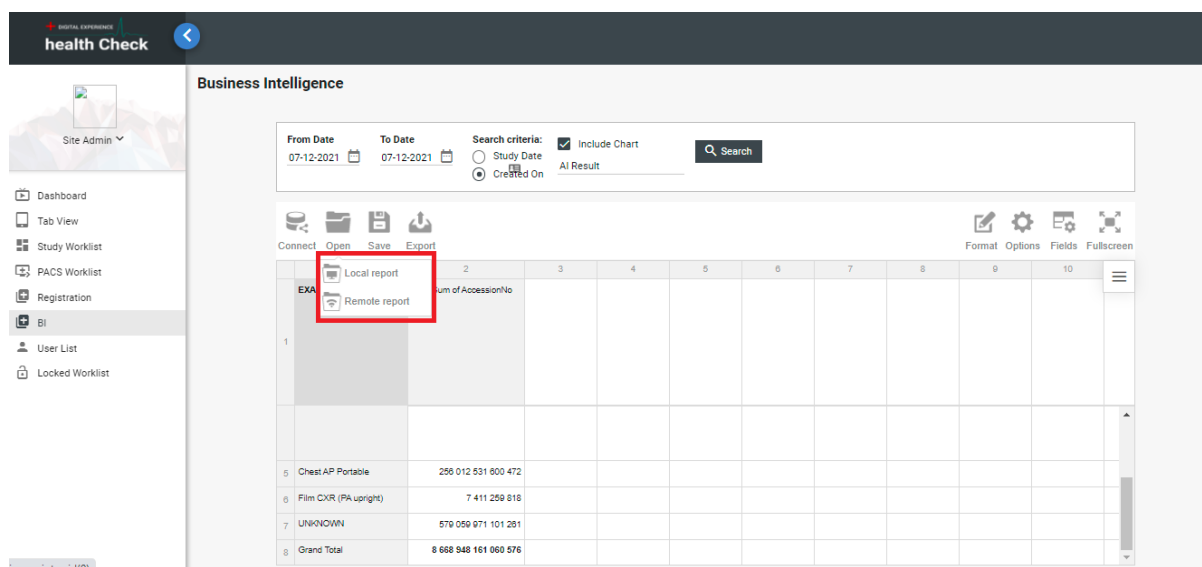
Connect

In this option, the user can connect any data from-

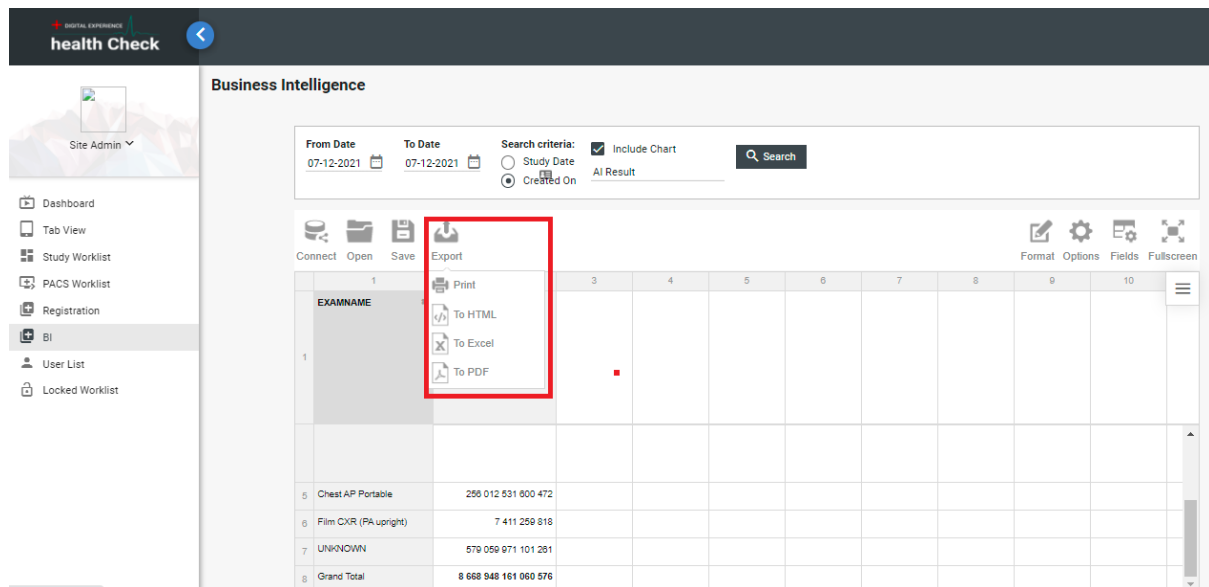
- To Local CSV
- To Local JSON
- To remote CSV
- To remote JSON



They can also open Local Report or Remote Report from the **Open** option.

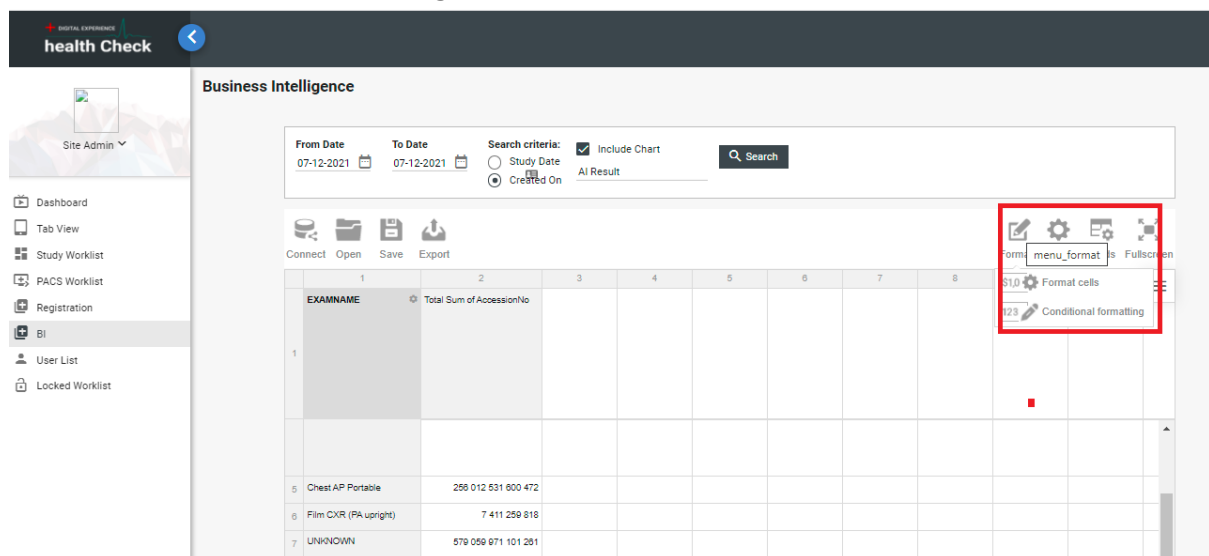


The user can also **Print**, export to HTML, Excel, and PDF format by clicking on the **Export** option.



In the right corner, the user will find **Format**, **Options**, **Fields**, and **Full-Screen** functions.

The user can easily format the data by clicking on the **Format**. The user will find **Format cells** and **Conditional formatting** inside the **Format** dropdown.



In the **Format cells**, the user will find some basic format changes options.

Format cells

APPLY

CANCEL

CHOOSE VALUE

Choose value

▼

Text align

right

▼

Thousand separator

(Space)

▼

Decimal separator

.

▼

Decimal places

None

▼

Currency symbol

Currency align

left

▼

Null value

Format as percent

false

▼

In **Conditional formatting**, the user will first find a pop-up window where they have to choose the ‘+’ button for further proceeding.

Conditional formatting

+

APPLY

CANCEL

There they can choose the format either less than, less than, or equal, and so on. Moreover, they can also choose a color for those format changes.

Conditional formatting

+

APPLY

CANCEL

Value:

All values

▼

Less than

▼

0

Format:

Arial

▼

12px

▼

A

73.93

×

The user can drag and drop fields to arrange the data inside the **Fields** option.

Fields

Add calculated value

APPLY

CANCEL

Drag and drop fields to arrange

All Fields
Expand All

> CreatedOn
> Dob
> OrderDate
☐ AbnormalityScore
☒ AccessionNo
☐ AtelectasisScore
☐ AtelectasisText
☐ CalcificationScore

Report Filters
Drop field here

Columns
Σ Values
Drop field here

Rows
ExamName
Drop field here

Values
Sum of AccessionNo Σ ∨
Drop field here

Note: The user can **Add calculated value** from the **Fields**.
Finally, they can maximize the screen with the **Fullscreen** option.

5. User List

You will find the details of User lists from here. Moreover, the user can update any users from the list. Also, they can add new users from the **User List**.

User List						
						+ Add New User
6 columns selected Global filter						
Full Name	Email	Mobile No	Created On	Status	Role Name	Action
Wariya Chintanapakdee	cwariya@hotmail.com	128		Approved	Radiologist	Update
Chomphunut Vijitsanguan	Chompoovj@gmail.com	818696369		Approved	Radiologist	Update
Wanwarang Teerasamit	Mojamp@gmail.com	0896986189		Approved	Radiologist	Update
Jitladda Wasinrat	jitladda_siriraj@hotmail.com	818265342		Approved	Radiologist	Update
Chalida Aphinives	achali@kku.ac.th	0857583636		Approved	Radiologist	Update

The user will get a form after clicking the **Add New User**.

User Information Entry

First Name: *

First Name

Last Name: *

Last Name

Phone No: *

Phone No

Email: *

Email

Password: *

Password

Confirm Password: *

Confirm Password

User Role: *

Select Role

Create

Create & Approve

Cancel

Note: * Mandatory Fields

6. Locked Worklist

The user will get the locked study from the **Study Worklist**.

Locked Worklist

Selected Group(s):

135 items selected

Search

7 columns selected

Global filter

☐	Locked By	Locked On	Accession#	HN	Patient Name	Exam Name	Status	Action

10

7. Tab View

The user can see the Tab View from here. There is an Auto-Refresh button that helps to refresh the lists.

ROUTER_AE2

Tuesday, Dec 07 2021

Auto Refresh

Refresh

in 60 sec

Close

NEXT PATIENTS

Queue

First Name

Last Name

8. Dashboard

In the dashboard, you can select Bus from the dropdown and see the **Patient Imaging Queue**. You can see both **Queue** and **Status** from the dashboard.

ROUTER_AE2

Bus :

MD-4 

Date : Dec 7, 2021

Patient Imaging Queue

Queue 

Status 